Panduan Sinkronisasi Waktu dengan NTP Meinberg

Observatorium Bosscha

2 April 2026

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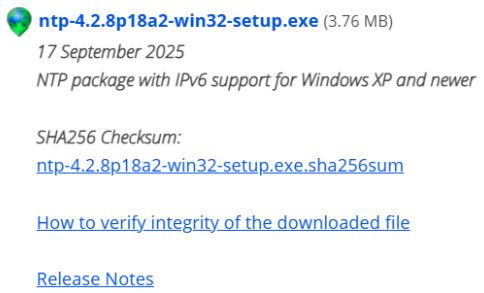
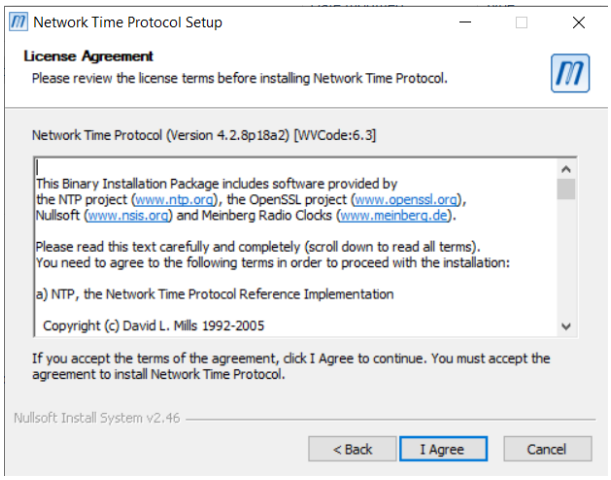
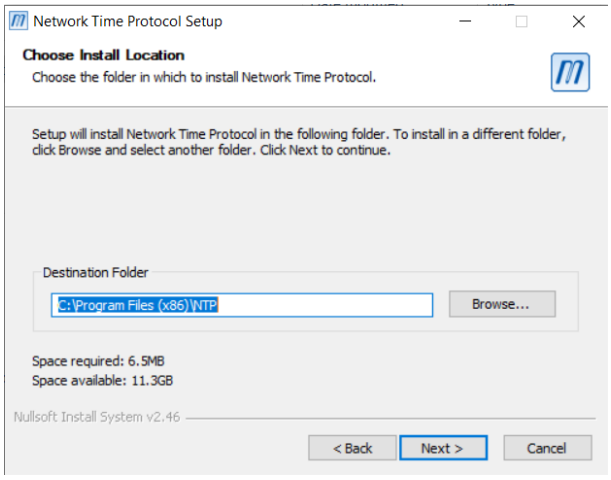
1. Unduh file instalasi NTP Meinberg dari alamat berikut: https://www.meinbergglobal.com/english/sw/ntp.htm



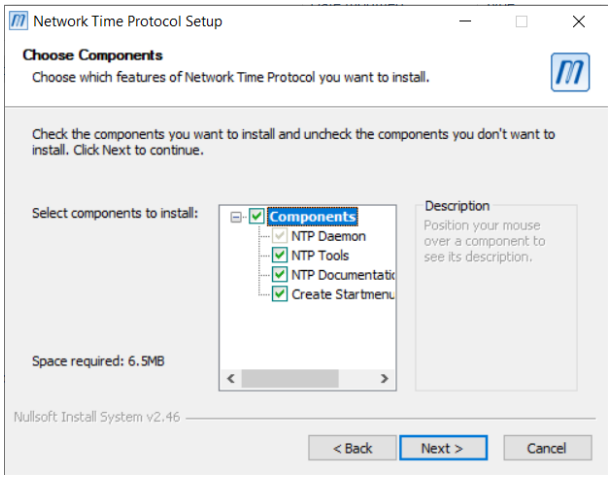
2. Klik ganda pada file, pilih **I Agree**.



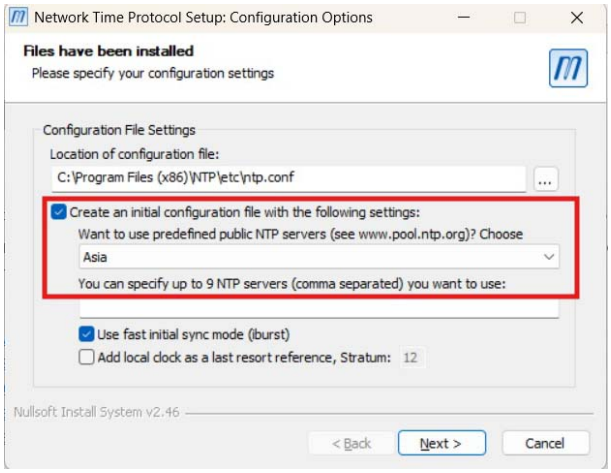
3. Tekan **Next**. Pada **Destination Folder** silakan tulis direktori tujuan instalasi, atau pilih default.



4. Centang semua pilihan. Tekan **Next**.

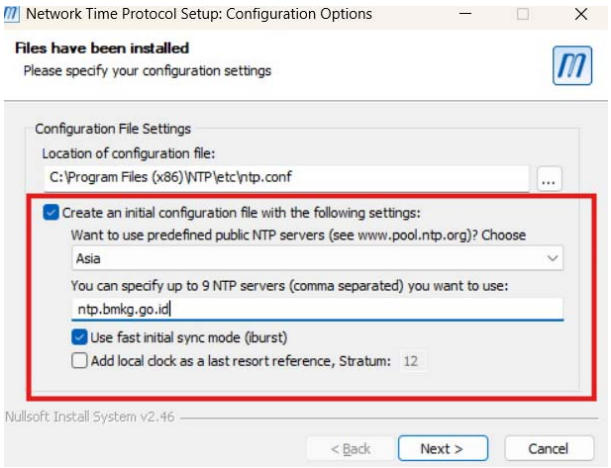


5. 'Want to use predefined...' pilih **Asia**.

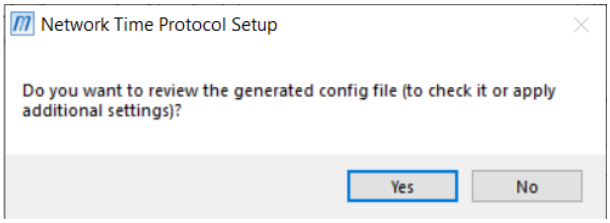


6. Isi bagian di dalam kotak merah dengan NTP server pilihan anda (contoh: `ntp.bmkg.go.id`).

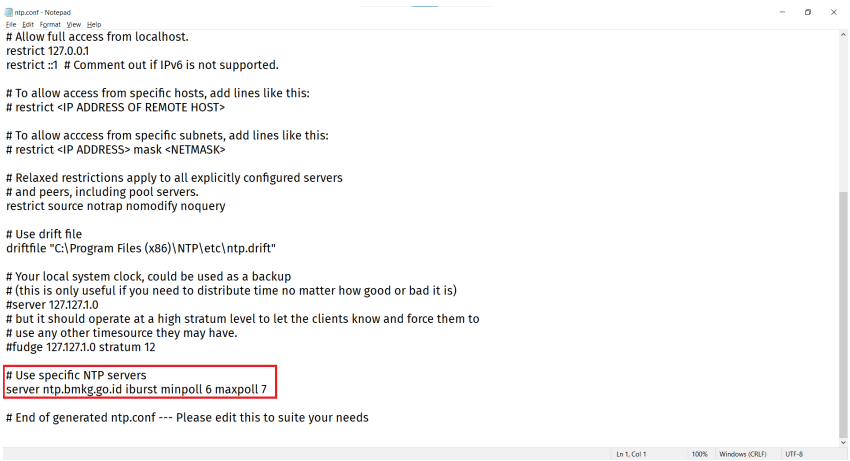
   File konfigurasi ini bisa diedit terpisah nantinya pada direktori default: `ProgramFiles(x86)/NTP/etc/ntp.conf` (sesuaikan dengan direktori instalasi anda). Tekan **Next**.



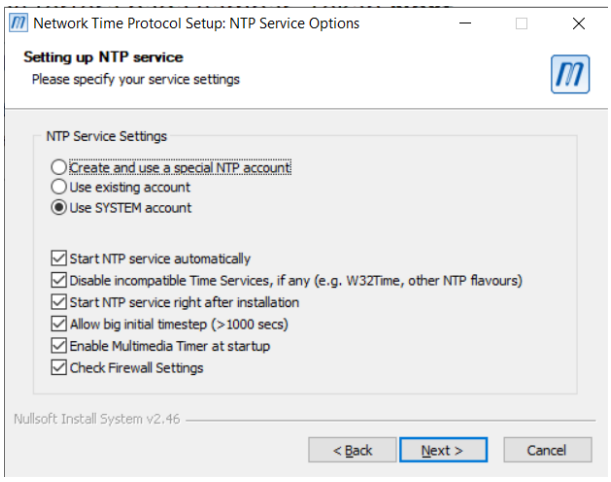
7. Tekan **Next**. Jendela verifikasi config file akan muncul. Pilih **Yes** jika ingin melihat isi filenya, atau **No** untuk melihat dan mengeditnya di kesempatan yang berbeda.



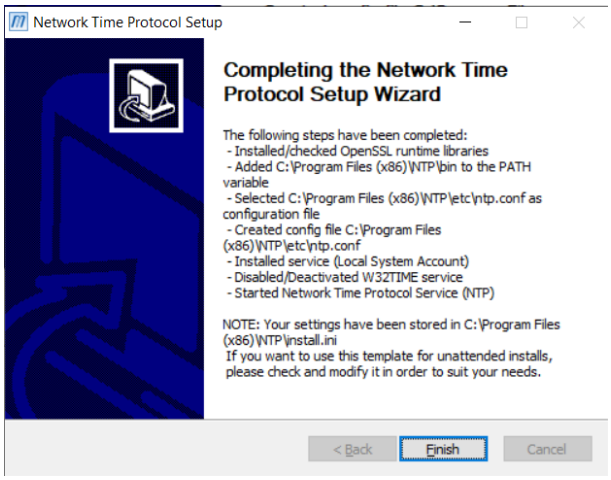
8. Jika memilih **Yes**, jendela berikut akan terbuka. NTP server yang dimasukkan pada poin 5 di atas akan tampil di file ini (lihat kotak merah). Copy dan Paste baris tersebut jika ingin menambahkan beberapa server lain. File config ini bisa diakses melalui direktori instalasi (lihat poin 5) dan mengeditnya dengan hak admin.



9. Sesuaikan seperti pada gambar. Tekan **Next**.



10. Tunggu instalasi selesai. Tekan **Finish**.



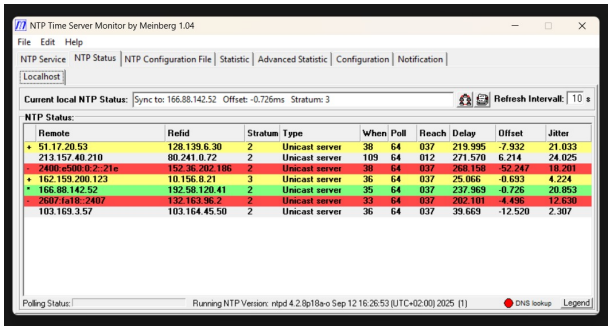
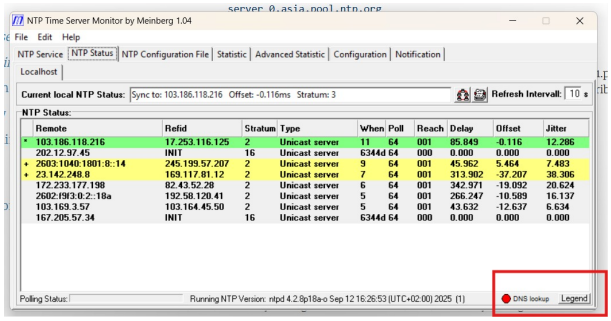
Catatan: Jika ingin menambahkan server NTP, edit file config pada mode **administrator** (lihat detil pada poin 7). Data server NTP, salah satunya, bisa dilihat pada https://www.ntppool.org/en/ dan pilih zona yang diinginkan (misal Asia). Berikut adalah contoh penggunaan multi server pada file config.

```
# NTP Network Time Protocol
# **** ATTENTION ****: *You have to restart the NTP service when you change this file to activate the changes*
# PLEASE CHECK THIS FILE CAREFULLY AND MODIFY IT IF REQUIRED
# Configuration File created by Windows Binary Distribution Installer Rev.: 1.27  mbg
# please check http://www.ntp.org for additional documentation and background information
# restrict access to avoid abuse of NTP for traffic amplification attacks
# see http://news.meinberg.de/244 for details
restrict default noquery nopeer nomodify notrap
restrict -6 default noquery nopeer nomodify notrap

# allow status queries and everything else from localhost
restrict 127.0.0.1
restrict -6 ::1

# if you need to allow access from a remote host, you can add lines like this:
# restrict <IP OF REMOTE HOST>

# Use drift file
driftfile "C:\Program Files (x86)\NTP\etc\ntp.drift"

# your local system clock, could be used as a backup
# (this is only useful if you need to distribute time no matter how good or bad it is)
#server 127.127.1.0
# but it should operate at a high stratum level to let the clients know and force them to
# use any other timesource they may have.
#fudge 127.127.1.0 stratum 12

# Use a NTP server from the ntp pool project (see http://www.pool.ntp.org)
# Please note that you need at least four different servers to be at least protected against
# one falseticker. If you only rely on internet time, it is highly recommended to add
# additional servers here.
# The 'iburst' keyword speeds up initial synchronization, please check the documentation for more details!
server 0.asia.pool.ntp.org iburst minpoll 6 maxpoll 7
server 1.asia.pool.ntp.org iburst minpoll 6 maxpoll 7
server 2.asia.pool.ntp.org iburst minpoll 6 maxpoll 7
server 0.us.pool.ntp.org iburst minpoll 6 maxpoll 7
server 1.us.pool.ntp.org iburst minpoll 6 maxpoll 7
server 2.us.pool.ntp.org iburst minpoll 6 maxpoll 7
```

11. Unduh dan instal NTP Client Monitor untuk memantau status sinkronisasi waktu. Aplikasi ini bisa diunduh dari https://www.meinbergglobal.com/download/ntp/windows/time-server-monitor/ntp-time-server-monitor-104.exe.

12. Buka aplikasi NTP Client Monitor pada mode **Administrator**.

    Masuk ke tab **NTP_Status** untuk melihat status sinkronisasi waktu.



13. Tekan tombol **legend** untuk melihat arti dari setiap status sinkronisasi. Status 'synchronized' dengan nilai *offset* yang kecil (idealnya di bawah 1 ms) menunjukkan bahwa waktu komputer sudah tersinkronisasi dengan baik ke server NTP.